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United States Patent	12411668
Kind Code	B2
Date of Patent	September 09, 2025
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Code generation tool for cloud-native high-performance computing

Abstract

Methods, storage media, and systems for translating a software expression from a user application programming interface (API) call to an API call of a software development kit (SDK) are disclosed. Some examples may include: receiving a tagged expression indicating that a translation of the software expression from a user API call to an API call of an SDK is to be performed, the SDK being associated with a cloud-native high-performance computing environment, processing an abstract syntax tree associated with the software expression, the processing including replacing symbols in the abstract syntax tree with respective variables, replacing a return statement in the abstract syntax tree with a serialization instruction to write a result to local storage, and serializing the processed abstract syntax tree and providing the serialized abstract syntax tree and one or more resource files to the cloud-native high-performance computing environment for execution.

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Appl. No.:	18/333517
Filed:	June 12, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230385039 A1	Nov. 30, 2023

Related U.S. Application Data

continuation parent-doc US 17321137 20210514 US 11714614 child-doc US 18333517
us-provisional-application US 63150146 20210217

Publication Classification

Int. Cl.: G06F8/40 (20180101); G06F8/30 (20180101); G06F9/48 (20060101)

U.S. Cl.:

CPC G06F8/40 (20130101); G06F8/30 (20130101); G06F9/48 (20130101);

Field of Classification Search

CPC: G06F (8/30); G06F (8/40); G06F (9/48)

USPC: 717/136; 717/140-144

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present disclosure is a continuation of and claims priority to U.S. patent application Ser. No. 17/321,137, entitled “CODE GENERATION TOOL FOR CLOUD-NATIVE HIGH-PERFORMANCE COMPUTING,” filed May 14, 2021, which claims priority to U.S. Provisional Patent Application No. 63/150,146 filed Feb. 17, 2021, the disclosures of which are hereby incorporated by reference in their entireties.

BACKGROUND

(1) Large-scale data-parallel optimization problems represent one of the most computationally challenging sets of scientific computing problems due to the high dimensionality of the unknown models and the large amounts of data involved. Traditionally, these workloads are deployed to on-premise high-performance computing (HPC) clusters and take advantage of massively shared file systems, fast inter-node connections, and highly resilient resources. The default approach for running these applications in the cloud is to replicate on-premise HPC clusters with on-demand infrastructure as a service (IaaS) resources such as HPC virtual machines and parallel file systems.

These on-premise and cloud-based cluster solutions come with high cost, low resilience, and increased management overhead. While some implementation services make cluster management and deployment more convenient, users are still responsible for selecting the right set of resources and managing the cluster throughout the cluster's lifetime. A large burden is placed on the user, who is often a domain-specialist without extensive HPC knowledge.

(2) Aside from the administrative complexity, on-demand clusters also suffer from shortcomings related to resilience and cost. While cluster management tools support elasticity and manage resilience, most applications running on clusters utilize a message passing interface (MPI) and do not support elasticity themselves. Unless an application has been specifically implemented to handle resilience, it cannot actually take advantage of resilience provided by a cloud platform. A similar limitation is true in terms of cost. Data-parallel optimization algorithms typically synchronize at a certain times during execution (e.g. to broadcast updates to all worker nodes), which leads to computational resources sitting temporarily idle. In principle, cloud services provide auto-scaling capabilities, but this is typically not supported by the application itself running on a cluster. The auto-scaling capabilities of services are therefore mainly used to scale resources in-between jobs and not to scale resources within an individual job itself.

(3) An alternative for running HPC workloads in the energy space on dedicated (virtual) clusters, are serverless function as a service (FaaS) and semi-serverless approaches that include a combination of FaaS and platform as a service (PaaS). Objective functions of data-parallel optimization problems like seismic imaging are embarrassingly parallel to evaluate and therefore present a great opportunity to leverage batch processing tools. Because iterative optimization algorithms also involve a serial component (i.e. collecting the updates, updating weights/model parameters), batch processing can be combined with workflow management tools that can express and execute directed acyclic graphs (DAGs). This can be achieved through serverless services, which effectively replace a dedicated master node. Using a combination of batch processing and serverless computations offers several advantages for users, such as inherent resilience (as tasks of a batch job are processed independently), the possibility to add or remove nodes during runtime, leverage resource harvesting, and virtually unlimited scalability (as no synchronization between worker nodes and a master node is required). These advantages translate to considerable cost savings for users, as serverless approaches can maximize resource utilization and thus reduce some costs by up to ninety percent. Serverless workflow management furthermore removes the classic master nodes as a single point of failure and enables automatic resource allocation for the workflow execution.

(4) While running large-scale data-parallel applications in a serverless and semi-serverless fashion provides many advantages, it requires that applications be fundamentally re-designed and re-implemented using multiple sets of potentially complex software development kits (SDKs). Furthermore, implementations become heavily platform-dependent and are not portable to other clouds or on-premise clusters. Having to implement serverless and semi-serverless approaches manually puts a large burden on the user and makes this approach not feasible for research and development purposes.

(5) It is with respect to these and other general considerations that embodiments have been described. Also, although relatively specific problems have been discussed, it should be understood that the embodiments should not be limited to solving the specific problems identified in the background.

SUMMARY

(6) Aspects of the present disclosure are directed to a framework for providing distributed extensions that allow users to offload computations to the cloud in a serverless and semi-serverless fashion based on batch and serverless computing. More specifically, such a framework can be realized by extending packages for distributed computing in high-level languages like Python or Julia to cloud services such as but not limited to AWS/Azure Batch and Azure (Durable) Functions,

AWS Lambda Functions or Google Cloud Functions. An additional layer of software abstraction is implemented on top of existing RESTful SDKs that allow users to offload computations to the cloud using remote procedure calls that resemble APIs for current (cluster-based) distributed packages. This enables users to run their code cloud-natively with minimal modifications to existing code using simple high-level code statements.

(7) In accordance with examples of the present disclosure, users can efficiently run data-parallel applications using batch processing and serverless computations, enabling them to scale to large-scale problem sizes across multiple regions and reduce cost by up to ninety percent. Thus, users can implement a single version of a program that can be executed on a local PC, a dedicated cloud or on-premises cluster, and cloud-natively using various PaaS and FaaS offerings. Thus, the gap between the development and deployment stage can be narrowed, as the same program can be executed in different computing environments. In some implementations, a user can interact with the cloud in a fully serverless fashion, where users can offload any type of computations to the cloud without having to specify or manage resources and, equally important, without having to make significant modifications to existing code.

(8) In accordance with aspects of the present disclosure, a method for translating a software expression from a user application programming interface (API) call to an API call of a software development kit (SDK) is described. The method may include receiving a tagged expression indicating that a translation of the software expression from a user API call to an API call of an SDK is to be performed, the SDK being associated with a cloud-native high-performance computing environment; processing an abstract syntax tree associated with the software expression, the processing including replacing symbols in the abstract syntax tree with respective variables, replacing a return statement in the abstract syntax tree with a serialization instruction to write a result to local storage, and serializing the processed abstract syntax tree; and providing the serialized abstract syntax tree and one or more resource files to the cloud-native high-performance computing environment for execution.

(9) In accordance with aspects of the present disclosure, a computer-readable storage medium including instructions, which when executed by a processor, cause the processor to translate a software expression from a user application programming interface (API) call to an API call of a software development kit (SDK) is described. The instructions, when executed by the processor, may cause the processor to: receive a tagged expression indicating that a translation of a software expression from a user API call to an API call of an SDK is to be performed, the SDK being associated with a cloud-native high-performance computing environment; process an abstract syntax tree associated with the software expression by replacing symbols in the abstract syntax tree with respective variables, replacing a return statement in the abstract syntax tree with a serialization instruction to write a result to local storage, and serializing the processed abstract syntax tree; and provide the serialized abstract syntax tree and one or more resource files to the cloud-native high-performance computing environment for execution.

(10) In accordance with aspects of the present disclosure, a system is described. The system may include one or more hardware processors configured by machine-readable instructions to: receive a tagged expression indicating that a translation of a software expression from a user API call to an API call of an SDK is to be performed, the SDK being associated with a cloud-native high-performance computing environment; process an abstract syntax tree associated with the software expression, the processing including replacing symbols in the abstract syntax tree with respective variables, replacing a return statement in the abstract syntax tree with a serialization instruction to write a result to local storage, and serializing the processed abstract syntax tree; and provide the serialized abstract syntax tree and one or more resource files to the cloud-native high-performance computing environment for execution.

(11) This summary is provided to introduce a selection of concepts in a simplified form that is further described below in the Detailed Description. This summary is not intended to identify key

features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Non-limiting and non-exhaustive examples are described with reference to the following Figures.
- (2) FIG. 1 depicts details of an operating environment for a code generation tool for a cloud-native high-performance computing environment in accordance with examples of the present disclosure.
- (3) FIG. 2 depicts details directed to a code generation tool in accordance with examples of the present disclosure.
- (4) FIG. 3 depicts details of a first macro expansion in accordance with examples of the present disclosure.
- (5) FIG. 4 depicts details of a second macro expansion in accordance with examples of the present disclosure.
- (6) FIG. 5 depicts details of a third macro expansion in accordance with examples of the present disclosure.
- (7) FIG. 6 depicts details of a method for generating a code for use at a cloud-native high-performance computing environment in accordance with examples of the present disclosure.
- (8) FIG. 7 depicts details of a method for generating a serialized abstract syntax table for use at a cloud-native high-performance computing environment in accordance with examples of the present disclosure.
- (9) FIG. 8 depicts details of a method for generating an environment for use at a cloud-native high-performance computing environment in accordance with examples of the present disclosure.
- (10) FIG. 9 depicts a block diagram illustrating physical components of a computing system with which aspects of the disclosure may be practiced.
- (11) FIGS. 10A-10B depict details of one or more computing systems in accordance with examples of the present disclosure.
- (12) FIG. 11 depicts an architecture of a system for processing data received at a computing system in accordance with examples of the present disclosure.

DETAILED DESCRIPTION

(13) In the following detailed description, references are made to the accompanying drawings that form a part hereof, and in which are shown by way of illustrations, specific embodiments, or examples. These aspects may be combined, other aspects may be utilized, and structural changes may be made without departing from the present disclosure. Embodiments may be practiced as methods, systems, or devices. Accordingly, embodiments may take the form of a hardware implementation, an entirely software implementation, or an implementation combining software and hardware aspects. The following detailed description is therefore not to be taken in a limiting sense, and the scope of the present disclosure is defined by the appended claims and their equivalents.

(14) The traditional approach to distributed computing for scientific applications is based on two-sided communication, mainly via the message passing interface (MPI) standard. Many existing legacy programs and coding applications, for example in the energy space, are built on this paradigm and implemented in lower-level languages like Fortran or C. While the MPI standard offers very fine-grained control over communication patterns, it comes with a high level of complexity that often stifles innovation and makes code maintenance cumbersome. With the rise of higher-level programming languages like Python or Julia, one-sided communication has become the de-facto standard for parallelizing task-parallel applications. Unlike two-sided communication,

users need to manage only the master process and utilize remote workers through remote function/procedure calls (RPCs).

(15) In the cloud, distributed applications based on one-sided communication can be executed on a cluster of interconnected virtual machines (VMs). However, if users want to leverage cloud-native services for (distributed) computing like batch processing or serverless functions (FaaS), this programming model no longer applies, as PaaS and FaaS offerings are almost exclusively exposed via REST APIs. Users can interact with these services through SDKs provided by the cloud platform, which are thus platform-specific, even though the underlying services are typically not. Therefore, if users want to run distributed applications using PaaS and FaaS offerings, they manually need to prepare their application and implement glue code for the specific cloud SDK. Glue code serves solely to “adapt” different parts of code that would otherwise be incompatible. Glue code does not contribute any functionality towards meeting program requirements; instead, glue code often appears in code that lets existing libraries or programs interoperate, as in language bindings or foreign function interfaces. For example, to run code as a batch job on the Microsoft Azure cloud, users need to create a pool of batch workers (each with their correct dependencies installed), prepare their application, create a batch job with the correct input and output bindings and specify how the remote workers execute the code. A simple hello-world-style example for Azure Batch would require almost 400 lines of Python code. Similar steps apply to running code serverless via Azure or AWS Lambda functions. As the required glue code is both heavily application and platform dependent, it oftentimes needs to be re-written multiple times.

(16) Existing solutions for distributed computing in the cloud fall into three main categories: (1) Cluster-based cloud computing, (2) Function and Platform as-a-service(s), and (3) Academic solutions.

(17) In examples, cluster-based cloud computing includes classic cluster managers (e.g. Azure CycleCloud, AWS ParallelCluster) or cluster managers for container orchestrations (e.g., Kubernetes, Docker Swarm). In either case, users are responsible for managing the life cycle of a cluster, which includes the creation of a set of instances, the establishment of a virtual network and connections between nodes/containers and the mounting of parallel file systems. Users are also (in their application) responsible for distributing and scheduling the parallel tasks within their program (e.g., as a parallel loop). This creates a large amount of management overhead for the user. Another disadvantage of the classic cluster approach for HPC in the cloud is the master-worker scheme on which cluster-based computing is based. This exposes the master-worker as a single point of failure and makes long-running applications prone to resilience problems.

(18) In examples, function and platform-as-a-service(s) includes serverless solutions (e.g., AWS Lambda, Azure Functions, Google Cloud Functions), managed compute and orchestration services (e.g., AWS Step Functions, Azure Durable Functions, AWS/Azure Batch, Azure Logic Apps) and cloud-native storage solutions (e.g., AWS S3, Azure Blob Storage, Dynamo DB, Cosmos DB). These services provide many advantages for running HPC workloads in the cloud, such as automated and semi-automated resource allocation, high scalability, and lower cost due to improved resource utilization. However, integrating these services into existing HPC applications requires major changes to the software, as cloud services are exposed via platform/vendor-specific APIs or require a specific representation of an application (e.g., a state machine expressed in JSON format). However, the software in HPC is based on two-side (MPI) or one-sided communication statements (i.e., tasks/futures/broadcast/reduce) that fundamentally differ from REST APIs of cloud services. Currently, there exists no commercial or academic solution that bridges this gap.

(19) Existing academic solutions tend to be directed to adapting serverless (FaaS) computing for general-purpose computations. Several frameworks enable users to execute workloads through serverless environments such as AWS Lambda, including PyWren, gg, mu, and Kappa. PyWren, gg, and mu are three academic serverless frameworks that enable running applications via a swarm of serverless functions. However, these projects are solely focused on FaaS and do not allow

executions through additional services (e.g., batch processing via AWS/Azure Batch). These frameworks require that user applications be manually partitioned into components that fit into the serverless function framework, i.e., they can be executed within given time and memory limits. Additionally, none of these existing frameworks provide a concurrency API that closely resembles existing APIs for cluster-based distributed programming.

(20) The Kappa framework is a serverless orchestration framework that does provide a concurrency API based on task-based distributed programming. However, Kappa executes workloads only through serverless functions and does not provide users access to other PaaS offerings. This severely limits the possibility of applying Kappa to real-world HPC applications due to serverless computing's hardware restrictions (limited memory/execution time). Kappa addresses the limited execution time through automatic checkpointing, but this is not a feasible solution to long-running HPC applications like seismic imaging, which often requires that several gigabytes or terabytes of data be frequently moved to and from storage. Technically, Kappa is implemented through checkpointing and captures the current state of a program using continuations.

(21) Thus, unlike existing cluster-based cloud computing techniques, examples provided herein allow users to take advantage of FaaS/PaaS offerings that shift responsibilities, such as being responsible for distributing and scheduling parallel tasks within programs (e.g., as a parallel loop), from the user to the cloud platform. Further, unlike Kappa, examples provided herein provide more flexibility to interface a broader range of services (e.g., for batch computing) and allow for creating IO bindings or resource files required for specific cloud services.

(22) Examples of the present disclosure are directed to utilizing distributed computing with cloud-native PaaS and FaaS offerings based on an additional layer of abstraction that sits on top of existing RESTful cloud SDKs. An API can be utilized that, at the user side, resembles existing APIs for distributed computing based on one-sided communication and remote function calls. On the back end, the remote function calls (RFCs) are translated to API calls for the various cloud services. For users, the additional abstraction level makes it easier to adapt cloud-native services for distributed computing and does not require any platform-specific glue code based on multiple cloud SDKs. These abstractions will therefore increase the productivity of users and open up PaaS and FaaS offerings to researchers and developers who want to focus on their application rather than on cloud architectures and how to deploy their code. Additionally, by being able to easily access services that otherwise require large amounts of glue code, users can leverage services that allow them to significantly reduce cost, increase resilience and scalability in comparison to cluster-based approaches.

(23) Examples of the present disclosure implement a software design based on a mullet architecture, i.e., an API that on the user side provides one-sided communication statements and RFCs, while the backend is implemented through RESTful cloud SDKs. Users are therefore able to write code that resembles conventional distributed applications, but remote function calls are executed via services such as batch computing or cloud functions. As these services do not provide direct networking connectivity between the user/master and workers/functions, leveraging traditional RPC APIs is not possible. As cloud services are almost always exposed via REST APIs based on HTTP endpoints, an abstraction layer is implemented that translates remote function calls and high-level code statements at the user side to REST API calls via existing cloud SDKs.

(24) In accordance with examples of the present disclosure, the translation of a remote function call to a cloud API call (including the creation of correct input-output bindings) is to be fully automatic. Therefore there is a need to translate a conventional function definition and function call to code that can be executed through a cloud SDK. For example, if a user defines a function with a return statement and then executes this function remotely, the expression can be analyzed, and the return statement can be replaced with a serialization step that writes the result to the local (remote) disk. A resource binding that automatically uploads the file to cloud object storage is implemented, making the data available to the user through another function call, without requiring the user to directly

interact with the storage client. The automatic code generation includes several steps, such as the capturing of input-output arguments, the creation of IO bindings, and the splitting of expressions into individual function calls or batch tasks. The generated code, along with the IO bindings, is then executed via cloud services using the respective SDK.

(25) In examples, the proposed framework can be implemented in the Julia language, a programming language that was designed from the ground up for numerical scientific and distributed computing. Similar to Python, Julia allows programming in a high-level fashion, enabling quick prototyping and development. However, unlike Python, Julia naturally supports metaprogramming, that is, expressions are represented through Julia data structures and can be analyzed and manipulated in the language itself. Additionally, and unlike Python, Julia offers optional typing and is based on just-in-time (JIT) compilation using LLVM for example, making the language nearly as fast as C code. Julia also interacts well to other programming languages, as it allows for a direct interface with Python, Fortran or C without any glue code.

(26) Julia also provides a built-in package for distributed computing, which is part of the language itself. Accordingly, one-sided remote function calls to cloud-based services can be enabled through extensions of Julia's distributed capabilities and provide a set of high-level instructions for executing functions, such as but not limited to AWS/Azure Batch jobs, Azure/Lambda/Google Cloud Functions, and Azure Durable Functions calls. Such functions enable users to define serverless workflows and execute batch computations via simple Julia macro statements. The Julia language provides such macros to map computations to a pool of inter-connected Julia workers and, in examples described herein, are extended to other cloud-based services, such as those services offered in an AWS, GCP or Azure environment. User-side remote function calls are translated to SDK calls to the respective cloud service by analyzing Julia code (i.e., expressions) and creating the required IO bindings to execute the code remotely. Although examples provided herein are generally directed to the Julia programming language, other programming languages are contemplated.

(27) FIG. 1 depicts details of an operating environment **100** for implementing a code generation tool for a cloud-native high-performance computing environment **124** in accordance with examples of the present disclosure. In examples, the operating environment **100** includes a computing device **104** at which a user may generate or otherwise create application instructions **110** defining a workload to be completed by the cloud-native high-performance computing environment **124**. The computing device **104**, although depicted as a desktop computer, for example, may be any one of a portable or non-portable computing device. Examples of the computing device **104** include, but are not limited to, a virtual machine, a laptop, a desktop, or a server. The cloud-native high-performance computing environment **124** describes an environment for large-scale workloads that require a large number of cores, often numbering in the hundreds or thousands. Scenarios where the cloud-native high-performance computing environment **124** may be utilized include but are not limited to image rendering, fluid dynamics, financial risk modeling, oil exploration, drug design, and engineering stress analysis. In examples, the work to be processed can be split into discrete tasks, which can be run across many cores simultaneously. In general, each task is finite, where the task takes an input, does some processing, and produces an output. For some applications, tasks are independent and can run in parallel. In other cases, tasks are tightly coupled, meaning they must interact or exchange intermediate results.

(28) A development environment **106** may be executed by the computing device **104**; the development environment generally includes a combination of a text editor and a runtime implementation. The text editor allows the user **102** to write code, or application instructions **110**, for a specific application or workload. In examples, the development environment **106** includes a user application programming interface (API) and code generator **116**. The user API & code generator **116** translates a remote function call to a cloud API call specific to the cloud-native high-performance computing environment **124**. For example, the application instructions **110** may

include one or more expressions **112A-112C**. The one or more expressions **112A-112C** may be associated with a previously generated program or workload deployed to on-premise HPC clusters. (29) To configure the application instructions **110** for use with the cloud-native high-performance computing environment **124**, for example, one or more tags (e.g., @macro_) may be used to identify one or more expressions **112A-112C** that are to be translated for use with a software development kit (SDK) to access REST and RESTful endpoints **125** associated with the cloud-native high-performance computing environment **124**. In some examples, a user **102** manually adds a tag to those expressions **112A-112C** needing translation. In other examples, the tag may be automatically added to expressions **112A-112C** needing translation. For example, a tag may be automatically associated with one or more expressions **112A-112C** that match an expression or keyword for translation. Accordingly, based on the tag identifying an expression for translation, a conventional function definition and function call associated with the expression (e.g., **114A-114C**) is translated to code that can be executed through a cloud SDK to access REST and RESTful endpoints **125** associated with the cloud-native high-performance computing environment **124**. For example, if a user-defined expression includes a function with a return statement and such function is to be executed at a cloud-native high-performance computing environment **124**, the expression is analyzed, and the return statement is replaced with a serialization step that writes the result to a file at a local or remote storage area. A resource binding is then automatically created, where the resource binding uploads the file to a storage area (such as cloud object storage) that is available to the user using another function call, but without requiring the user to directly interact with a storage client.

(30) The automatic translation includes several steps such as capturing of input-output arguments, the creation of IO bindings is so required by the cloud service, and the splitting of expressions into individual function calls or batch tasks. As an example, abstract syntax trees are collected for each expression that is to be executed at the cloud-native high-performance computing environment **124**. An abstract syntax tree is a tree representation of the abstract syntactic structure of source code written in a programming language. Each node of the tree denotes a construct occurring in the source code. Each of the collected abstract syntax trees is analyzed to capture runtime arguments and to replace symbols of variables with actual variables. Each of the abstract syntax trees is then serialized and provided to the cloud-native high-performance computing environment **124** as a resource file via the network **140**. For expressions with return statements, the return statements are replaced with an output file name (e.g., randomly named output file). The output file name is then added as an output resource and provided to the cloud-native high-performance computing environment **124** via the network **140**.

(31) An example abstract syntax tree for an algorithm (while $b \neq 0$, $a := a - b$, return a) is provided as abstract syntax tree **160**. The abstract syntax tree **160** may include a statement sequence **162** and return a value **166** (e.g., a) indicated by a return statement **164**. A while branch **168** indicates that the while loop would continue to execute in accordance with a compare statement **170** (e.g., $\neq$) comparing a variable **174** (e.g., b) to a constant **176** (e.g., 0). An assignment at **172** would assign the output of the operation (e.g., “-”) between the variable **182** (e.g., a) and variable **184** (e.g., b) to the variable **178** (e.g., a). In examples, the symbols representing the variables (e.g., a and b) would be replaced with actual variables. The return statement (e.g., **164**) would be replaced with a serialization of the return argument (e.g., a) and a write operation of the serialized file to local disk storage or cloud object storage using a randomly generated file name. Accordingly, the processed abstract syntax tree can then be serialized and provided as one or more resource files **118**. The one or more resource files **118** may include the translated function call and/or the output file name.

(32) As depicted in FIG. 1, resources **120**, including input file(s) and application(s) to process the input file(s), may be uploaded to the cloud-native high-performance computing environment **124**. The input files can be any data that the application instructions process, such as financial modeling data, video files to be transcoded, etc. The application files can include scripts or applications, such

as the one or more resource files **118** that process the data. The resources **120** may be uploaded to a storage area **138**. A pool **134** of compute nodes **136A-136B** may be created and associated with a cloud services account **126**. The cloud services account **126** is associated with the cloud-native high-performance computing environment **124** and determines what computing resources are available and in what type of configuration such resources exist. A compute node **136** may be a virtual machine that processes a portion of an application's workload. For example, a compute node **136** may process or otherwise execute one or more tasks associated with a job **130**. Job **130** may be defined to run the workload on pool **134**, where pool **134** is a collection of nodes that the application is to run on. A job **130** is a collection of tasks **132A-132C**. Job **130** manages how computation is performed by its tasks **132A-132C** on the compute nodes **136A-136C** in pool **134**. A job **130** specifies the pool **134** in which the work is to be run. A new pool **134** can be created for each job **130**, or one pool **134** may be used for many jobs **130**. A task, such as task **132A**, is a computation unit associated with a job, such as job **130**. Each task **132A-132C** runs on a respective node **136A-136C**. Tasks **132** are assigned to nodes **136** for execution or are queued until a node **136** becomes free. Task **132** runs one or more programs or scripts on a compute node **136** to perform the work that is to be done. The tasks **132A-132C** may be defined per the resources **120** and associated with job **130**.

(33) FIG. 2 depicts a framework for distributed computing based on cloud Paas and Faas components in accordance with examples of the present disclosure. More specifically, the code generator **206** bridges a platform independent user API **204** and a cloud SDK to provide existing workloads and applications originally designed for use with HPC clusters access to cloud services **210**. The user API **204** can be platform-independent and provides remote function calls and high-level one-sided communication statements. The code generator **206** can be platform-independent and translates remote function calls, and high-level code statements at the user side (e.g., the user API **204**) to REST API calls via existing and platform-dependent cloud SDKs **208**. The user API **204**, code generator **206**, and the cloud SDK **208** may be implemented in a user computing device, such as the computing device **104** (FIG. 1) and/or a virtual machine. The cloud services **210** may be accessible via the REST and RESTful endpoints **125** (FIG. 1).

(34) FIG. 3 depicts a first example of a macro implemented at the user API **304** in accordance with examples of the present disclosure. The macro `@bcast` can be added to an existing expression to make the existing expression available as a resource at a cloud-native high-performance computing environment. The code generator **306**, which may provide the same as or similar functions as the user API & code generator **116** (FIG. 1), may translate the expression from the user API **304** to create an object storage container, upload the expression to cloud object storage, and create a batch resource file from the object using a cloud API call **308** exposed via a cloud SDK. More specifically, the code generator **306** may expand the expression into a macro that serializes the expression, generates the required resource bindings, and returns a future, or reference to object storage, such as a blob, to the user. Accordingly, implementing the `@bcast` expr macro via the user API **304** allows the existing expression `expr` to be implemented at a cloud-native high-performance computing environment.

(35) FIG. 4 depicts a second example of a macro implemented at the user API **404** in accordance with examples of the present disclosure. The macro `@batchdef` can be added to an existing expression to ready the expression for execution on the local machine and/or on subsequent batch workers and/or serverless functions. The code generator **406**, which may provide the same as or similar functions as the user API & code generator **116** (FIG. 1), appends the expression (its abstract syntax tree) to a global state variable. Upon a call to `@batchexec`, the code generator **406** may serialize the expression, and then upload the serialized expression to the cloud storage, such as Azure blob storage. Accordingly, the user API **404** and the code generator **406** do not make a call to the cloud API **408** until a call to `@batchexec` is made.

(36) FIG. 5 depicts a third example of a macro implemented at the user API **504** in accordance with

examples of the present disclosure. The macro `@batchexec` can be added to an existing expression to execute the existing expression as a single or multitask batch job in the cloud. In examples, the code generator **506**, which may provide the same as or similar functions as the user API & code generator **116** (FIG. 1) translates the expression from the user API **504** to upload serialized abstract syntax trees to the storage, create batch input/output resources or other resources required by the respective cloud service, and to create and submit jobs and tasks to the cloud-native high-performance computing environment using a cloud API call **508** exposed via a cloud SDK. More specifically, the code generator **506** expands the expression `expr` by collecting and analyzing all symbols in an abstract syntax tree. In examples, those expressions, including a parallel mapping (`pmap`), may be split into individual tasks such that a single expression is created for each task to be performed in parallel. The code generator **506** replaces the symbol variables in the abstract syntax tree with actual variables, replaces return statements in the abstract syntax tree with serialization, serializes the abstract syntax tree, and uploads the serialized syntax tree to the storage, such as Azure blob storage. In addition, the code generator **506** creates input/output bindings, including but not limited to input resource files and output resource files as previously described. In examples, the `@batchexec expr` creates jobs, such as batch jobs and tasks and may return a reference to future output stored in object storage. When executed at the cloud-native high-performance computing environment, a worker or serverless function may de-serialize and compile the abstract syntax tree into an executable which is then executed on the local hardware. The cloud API call **508** provides an interface to upload the serialized abstract syntax tree to storage, create input/output resources (depending on the cloud service that is being called), and then create and submit jobs/tasks for execution. Accordingly, implementing the `@batchexec expr` macro via the user API **504** allows the existing expression `expr` to be implemented at a cloud-native high-performance computing environment.

(37) The user APIs **304** (FIG. 3), **404** (FIG. 4), and **504** (FIG. 5) are the same user API providing multiple macro expansion functions. Similarly, the code generator **306** (FIG. 3), **406** (FIG. 4), and **506** (FIG. 5) are the same code generator translating expression from the user API to cloud API calls and/or cloud SDKs. The cloud API call **308** (FIG. 3), **408** (FIG. 4), and **508** (FIG. 5) are the same cloud API/SDK and are dependent upon the cloud-native high-performance computing environment.

(38) FIG. 6 depicts details of a method **600** for receiving, generating, and then executing code at a cloud-native high-performance computing environment in accordance with examples of the present disclosure. A general order for the steps of method **600** is shown in FIG. 6. Generally, method **600** starts at **602** and ends at **610**. Method **600** may include more or fewer steps or may arrange the order of the steps differently than those shown in FIG. 6. Method **600** can be executed as a set of computer-executable instructions executed by a computer system and encoded or stored on a computer-readable medium. In examples, aspects of the method **600** are performed by one or more processing devices, such as a computer or server. Further, the method **600** can be performed by gates or circuits associated with a processor, Application Specific Integrated Circuit (ASIC), a field programmable gate array (FPGA), a system on chip (SOC), a neural processing unit, or other hardware device. Hereinafter, the method **600** shall be explained with reference to the systems, components, modules, software, data structures, user interfaces, etc., described in conjunction with FIGS. 1-5.

(39) The method starts at **602**, where flow may proceed to **604**. At **604**, tagged code, including tagged application expressions such as but not limited to the application instructions **110** (FIG. 1), including tagged expressions **114A-114C** (FIG. 1), may be received and/or analyzed. For example, a development environment **106** (FIG. 1) may be executed by the computing device **104** (FIG. 1) and includes code, or application instructions **110** (FIG. 1), for a specific application or workload. In examples, the development environment **106** includes a user application programming interface (API) and code generator **116**. The expressions included in the code may be analyzed, and those

expressions, including a matching macro tag, may be identified. At **606**, each of the expressions having a macro tag may be expanded. In examples, the expansion of the macros may be performed by a code generator **306/406/506** and/or the user API & code generator **116**. The user API & code generator **116** translates a remote function call to one or multiple cloud API calls specific to the cloud-native high-performance computing environment **124**.

(40) Additionally, the macro expansion **606** may include configuring one or more parameters to manage the requirements for performing the computing operations. For example, this may include creating the batch job, creating a container in cloud object storage, and creating specialized input and output bindings that may be required by the respective cloud service (e.g. the batch service). Method **600** may proceed to **608**, where the batch job is executed at the cloud-native high-performance computing environment. Method **600** may return data or provide access to the data via one or more output resources in the form of remote references. Method **600** may end at **610**.

(41) FIG. 7 depicts a method **700** for translating remote function calls to API calls available via a cloud SDK in accordance with examples of the present disclosure. A general order for the steps of method **700** is shown in FIG. 7. Generally, method **700** starts at **702** and ends at **712**. Method **700** may include more or fewer steps or may arrange the order of the steps differently than those shown in FIG. 7. Method **700** can be executed as a set of computer-executable instructions executed by a computer system and encoded or stored on a computer-readable medium. In examples, aspects of method **700** are performed by one or more processing devices, such as a computer or server.

Further, method **700** can be performed by gates or circuits associated with a processor, Application Specific Integrated Circuit (ASIC), a field-programmable gate array (FPGA), a system on chip (SOC), a neural processing unit, or other hardware devices. Hereinafter, method **700** shall be explained with reference to the systems, components, modules, software, data structures, user interfaces, etc., described in conjunction with FIGS. 1-6.

(42) The method starts at **702**, where flow may proceed to **704**. At **704**, abstract syntax trees are collected for each expression to be executed in the cloud-native high-performance computing environment. Expression may represent function definitions/calls, variable assignments, load statements of modules and others. Each of the collected abstract syntax trees is analyzed to capture runtime arguments (in case of function calls), and symbols of variables are replaced with actual variables at **706**. Each of the abstract syntax trees is then serialized at **708** and provided to the cloud-native high-performance computing environment as a resource file at **719**. For expressions with return statements (i.e., function calls), the return statements are replaced in **706** with a serialization of the return argument and a write statement of the serialized expression to the local storage of the remote cloud worker using a randomly generated file/object name. Depending on the cloud service that is used to execute the computations, it may be required to create an output binding for the generated object, so that the locally stored object is moved to a durable cloud storage service upon the completion of the computations in **710**, from where it is accessible to the user at a later point in time. Method **700** ends at **712**.

(43) FIG. 8 depicts details of a method **800** for creating/configuring one or more parameters to manage the requirements for performing computing operations in accordance with examples of the present disclosure. A general order for the steps of method **800** is shown in FIG. 8. Generally, method **800** starts at **802** and ends at **810**. Method **800** may include more or fewer steps or may arrange the order of the steps differently than those shown in FIG. 8. The method **800** can be executed as a set of computer-executable instructions executed by a computer system and encoded or stored on a computer-readable medium. In examples, aspects of method **800** are performed by one or more processing devices, such as a computer or server.

(44) Further, method **700** can be performed by gates or circuits associated with a processor, Application Specific Integrated Circuit (ASIC), a field-programmable gate array (FPGA), a system on chip (SOC), a neural processing unit, or other hardware devices. Hereinafter, the method **800** shall be explained with reference to the systems, components, modules, software, data structures,

user interfaces, etc., described in conjunction with FIGS. 1-7.

(45) The method starts at **802**, where flow may proceed to **804**. At **804**, a cloud computing service is initiated which may include (but may not be limited to) cloud batch jobs or serverless function executions. In case of cloud batch jobs, the initiated job is a collection of tasks, where the cloud service manages how the set of parallel tasks are executed in a pool of compute nodes. The method may proceed to **806**, where a container for an application is created. Using containers provides a way to run tasks without having to manage an environment and dependencies to run applications. Containers deploy applications as lightweight, portable, self-sufficient units that can run in several different environments. Container-based tasks can also take advantage of features of non-container tasks, including application packages and management of resource files and output files. The method may proceed to **808**, where specialized resource storage containers are created. Resource files put data onto a virtual machine in the cloud-native high-performance computing environment, but the type of data and how it is used is flexible. In examples, there are a few options to generate resource files that may depend on where data is to be stored. Such options may include a storage container URL, a storage container name, and a web endpoint. The storage container URL generates a resource file from any storage container. The storage container name generates a resource file from the name of a container in a linked storage account. The web endpoint generates a resource file from any valid HTTP URL.

(46) The creation process for resource files varies depending on where the original data is stored. The specialized resource containers may include but are not limited to an input resource file and an output resource file. Method **800** may end at **810**.

(47) FIGS. 9-11 and the associated descriptions provide a discussion of a variety of operating environments in which aspects of the disclosure may be practiced. However, the devices and systems illustrated and discussed with respect to FIGS. 9-11 are for purposes of example and illustration and are not limiting of a vast number of computing system configurations that may be utilized for practicing aspects of the disclosure described herein.

(48) FIG. 9 is a block diagram illustrating physical components (e.g., hardware) of a computing system **900** with which aspects of the disclosure may be practiced. The computing system components described below may be suitable for the computing and/or processing devices described above. In a basic configuration, the computing system **900** may include at least one processing unit **902** and a system memory **904**. Depending on the configuration and type of computing system, the system memory **904** may comprise, but is not limited to, volatile storage (e.g., random-access memory (RAM)), nonvolatile storage (e.g., read-only memory (ROM)), flash memory, or any combination of such memories.

(49) The system memory **904** may include an operating system **905** and one or more program modules **906** suitable for running software application **920**, such as one or more components supported by the systems described herein. As examples, system memory **904** may include the user API **921**, the code generator **922**, and the cloud SDK **923**. The user API **921** may be the same as or similar to the user API **204** (FIG. 2), **304** (FIG. 3), **404** (FIG. 4), and **504** (FIG. 5). The code generator **922** may be the same as or similar to the code generator **2066** (FIG. 2), **306** (FIG. 3), **406** (FIG. 4), and **506** (FIG. 5). The cloud SDK may be the same as or similar to the cloud SDK **208** (FIG. 2), **308** (FIG. 3), **408** (FIG. 4), and **508** (FIG. 5). In examples, the user API & code generator **116** (FIG. 1) may include the user API **921** and the code generator **922**. The operating system **905**, for example, may be suitable for controlling the operation of the computing system **900**.

(50) Furthermore, embodiments of the disclosure may be practiced in conjunction with a graphics library, other operating systems, or any other application program and are not limited to any particular application or system. This basic configuration is illustrated in FIG. 9 by those components within a dashed line **908**. The computing system **900** may have additional features or functionality. For example, the computing system **900** may also include additional data storage devices (removable and/or non-removable) such as magnetic disks, optical disks, or tape. Such

additional storage is illustrated in FIG. 9 by a removable storage device **909** and a non-removable storage device **910**.

(51) As stated above, a number of program modules and data files may be stored in the system memory **904**. While executing on the processing unit **902**, the program modules **906** (e.g., software applications **920**) may perform processes including, but not limited to, the aspects as described herein. Other program modules that may be used in accordance with aspects of the present disclosure may include electronic mail and contacts applications, word processing applications, spreadsheet applications, database applications, slide presentation applications, drawing or computer-aided programs, etc.

(52) Furthermore, embodiments of the disclosure may be practiced in an electrical circuit, discrete electronic elements, packaged or integrated electronic chips containing logic gates, a circuit utilizing a microprocessor, or on a single chip containing electronic elements or microprocessors. For example, embodiments of the disclosure may be practiced via a system-on-a-chip (SOC) where each or many of the components illustrated in FIG. 9 may be integrated onto a single integrated circuit. Such a SOC device may include one or more processing units, graphics units, communications units, system virtualization units, and various application functionality, all of which are integrated (or “burned”) onto the chip substrate as a single integrated circuit. When operating via a SOC, the functionality described herein, with respect to the capability of the client to switch protocols, may be operated via application-specific logic integrated with other components of the computing system **900** on the single integrated circuit (chip). Embodiments of the disclosure may also be practiced using other technologies capable of performing logical operations such as, for example, AND, OR, and NOT, including but not limited to mechanical, optical, fluidic, and quantum technologies. In addition, embodiments of the disclosure may be practiced within a general-purpose computer or in any other circuits or systems.

(53) The computing system **900** may also have one or more input device(s) **912** such as a keyboard, a mouse, a pen, a sound or voice input device, a touch or swipe input device, etc. The output device(s) **914**, such as a display, speakers, a printer, etc., may also be included. The aforementioned devices are examples, and others may be used. The computing system **900** may include one or more communication connections **916**, allowing communications with other computing systems **950**. Examples of suitable communication connections **916** include, but are not limited to, radio frequency (RF) transmitter, receiver, and/or transceiver circuitry; universal serial bus (USB), parallel, and/or serial ports.

(54) The term computer-readable media as used herein may include computer storage media. Computer storage media may include volatile and nonvolatile, removable, and non-removable media implemented in any method or technology for storage of information, such as computer-readable instructions, data structures, or program modules. The system memory **904**, the removable storage device **909**, and the non-removable storage device **910** are all computer storage media examples (e.g., memory storage). Computer storage media may include RAM, ROM, electrically erasable read-only memory (EEPROM), flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage, or other magnetic storage devices, or any other article of manufacture which can be used to store information, and which can be accessed by the computing system **900**. Any such computer storage media may be part of the computing system **900**. Computer storage media does not include a carrier wave or other propagated or modulated data signal.

(55) Communication media may be embodied by computer-readable instructions, data structures, program modules, or other data in a modulated data signal, such as a carrier wave or other transport mechanism, and includes any information delivery media. The term “modulated data signal” may describe a signal that has one or more characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, communication media may include wired media such as a wired network or direct-wired connection, and wireless media such

as acoustic, radio frequency (RF), infrared, and other wireless media.

(56) FIGS. **10A-10B** illustrate a computing system **1000**, for example, a mobile telephone, a smart phone, wearable computer (such as a smart watch), a tablet computer, a desktop computer, a laptop computer, and the like, with which embodiments of the disclosure may be practiced. With reference to FIG. **10A**, one aspect of a computing system **1000** for implementing the aspects is illustrated. In a basic configuration, the computing system **1000** is a desktop computer having both input elements and output elements. The computing system **1000** typically includes a display **1005**, which may also function as an input device (e.g., a touch screen display). The computing system **1000** may also include a keypad **1035**. The keypad **1035** may be a physical keypad or a “soft” keypad generated on the touch screen display.

(57) In various embodiments, the output elements include the display **1005** for showing a graphical user interface (GUI), a visual indicator **1020** (e.g., a light-emitting diode), and/or an audio transducer **1025** (e.g., a speaker). In yet another aspect, the computing system **1000** incorporates input and/or output ports, such as an audio input (e.g., a microphone jack), an audio output (e.g., a headphone jack), and a video output (e.g., an HDMI port) for sending signals to or receiving signals from an external device.

(58) FIG. **10B** is a block diagram illustrating the architecture of one aspect of a mobile computing system. That is, the computing system **1000** can incorporate a system (e.g., an architecture) **1002** to implement some aspects. In one embodiment, system **1002** is implemented as a “computing system” capable of running one or more applications (e.g., browser, e-mail, calendaring, contact managers, messaging clients, games, and media clients/players). In some aspects, system **1002** is integrated as a computing system, such as a desktop computer.

(59) One or more application programs **1066** may be loaded into the memory **1062** and run on or in association with the operating system **1064**. Examples of the application programs include phone dialer programs, e-mail programs, personal information management (PIM) programs, word processing programs, spreadsheet programs, Internet browser programs, messaging programs, maps programs, and so forth. System **1002** also includes a nonvolatile storage area **1068** within the memory **1062**. The nonvolatile storage area **1068** may be used to store persistent information that should not be lost if the system **1002** is powered down. The application programs **1066** may use and store information in the nonvolatile storage area **1068**, such as e-mail or other messages used by an e-mail application, and the like. A synchronization application (not shown) also resides on system **1002** and is programmed to interact with a corresponding synchronization application resident on a host computer to keep the information stored in the nonvolatile storage area **1068** synchronized with corresponding information stored at the host computer. As should be appreciated, other applications may be loaded into the memory **1062** and run on the computing system **1000** described herein (e.g., search engine, extractor module, relevancy ranking module, answer scoring module, etc.).

(60) The system **1002** has a power supply **1070**, which may be implemented as one or more batteries. The power supply **1070** might further include an external power source, such as an AC adapter or a powered docking cradle that supplements or recharges the batteries.

(61) The system **1002** may also include a radio interface layer **1072** that performs the function of transmitting and receiving radio frequency communications. The radio interface layer **1072** facilitates wireless connectivity between the system **1002** and the “outside world” via a communications carrier or service provider. Transmissions to and from the radio interface layer **1072** are conducted under the control of the operating system **1064**. In other words, communications received by the radio interface layer **1072** may be disseminated to the application programs **1066** via the operating system **1064**, and vice versa.

(62) The system **1002** may further include a video interface **1076** that enables an operation of an on-board camera **1030** to record still images, video stream, and the like. A computing system **1000** implementing the system **1002** may have additional features or functionality. For example, the

computing system **1000** may also include additional data storage devices (removable and/or non-removable) such as magnetic disks, optical disks, or tape. Such additional storage is illustrated in FIG. **10B** by the nonvolatile storage area **1068**.

(63) Data/information generated or captured by the computing system **1000** and stored via the system **1002** may be stored locally on the computing system **1000**, as described above, or the data may be stored on any number of storage media that may be accessed by the device via the radio interface layer **1072** or via a wired connection between the computing system **1000** and a separate computing system associated with the computing system **1000**, for example, a server computer in a distributed computing network, such as the Internet. As should be appreciated, such data/information may be accessed via the computing system **1000** via the radio interface layer **1072** or via a distributed computing network. Similarly, such data/information may be readily transferred between computing systems for storage and use according to well-known data/information transfer and storage means, including electronic mail and collaborative data/information sharing systems.

(64) FIG. **11** illustrates one aspect of the architecture of a system for processing data received at a computing system from a remote source, such as a personal computer **1104**, a tablet computing device **1106**, or mobile computing device **1108**, as described above. Content at a server device **1102** may be stored in different communication channels or other storage types. For example, various documents may be stored using a directory service **1127**, a web portal **1128**, a mailbox service **1129**, an instant messaging store **1130**, or social networking services **1131**.

(65) One or more of the previously described program modules **906** or software applications **920** may be employed by server device **1102** and/or the personal computer **1104**, a tablet computing device **1106**, or mobile computing device **1108**, as described above. For example, the server device **1102** may include the user API **1121**, the code generator **1122**, and the cloud SDK **1123**. The user API **1121** may be the same as or similar to the user API **204** (FIG. **2**), **304** (FIG. **3**), **404** (FIG. **4**), **504** (FIG. **5**), and **921** (FIG. **9**). The code generator **1122** may be the same as or similar to the code generator **2066** (FIG. **2**), **306** (FIG. **3**), **406** (FIG. **4**), **506** (FIG. **5**), and **922** (FIG. **9**). The cloud SDK may be the same as or similar to the cloud SDK **208** (FIG. **2**), **308** (FIG. **3**), **408** (FIG. **4**), **508** (FIG. **5**), and **923** (FIG. **9**). In examples, the user API & code generator **116** (FIG. **1**) may include the user API **1121** and the code generator **1122**.

(66) The server device **1102** may provide data to and from a client computing device such as a personal computer **1104**, a tablet computing device **1106**, and/or a mobile computing device **1108** (e.g., a smart phone) through a network **1115**. By way of example, the computer system described above may be embodied in a personal computer **1104**, a tablet computing device **1106**, and/or a mobile computing device **1108** (e.g., a smart phone). Any of these embodiments of the computing devices may obtain content from the store **1116**, in addition to receiving graphical data useable to be either pre-processed at a graphic-originating system or post-processed at a receiving computing system.

(67) In addition, the aspects and functionalities described herein may operate over distributed systems (e.g., cloud-based computing systems), where application functionality, memory, data storage, and retrieval and various processing functions may be operated remotely from each other over a distributed computing network, such as the Internet or an intranet. User interfaces and information of various types may be displayed via on-board computing device displays or via remote display units associated with one or more computing devices. For example, user interfaces and information of various types may be displayed and interacted with on a wall surface onto which user interfaces and information of various types are projected. Interaction with the multitude of computing systems with which embodiments of the invention may be practiced includes keystroke entry, touch screen entry, voice or other audio entry, gesture entry where an associated computing device is equipped with detection (e.g., the camera) functionality for capturing and interpreting user gestures for controlling the functionality of the computing device, and the like.

(68) Aspects of the present disclosure, for example, are described above with reference to block

diagrams and/or operational illustrations of methods, systems, and computer program products according to aspects of the disclosure. The functions/acts noted in the blocks may occur out of order, as shown in any flowchart. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality/acts involved.

(69) The present disclosure relates to systems and methods for translating a software expression from a user application programming interface (API) call to an API call of a software development kit (SDK) according to at least the examples provided in the sections below: (A) In one aspect, some examples include a method for translating a software expression from a user application programming interface (API) call to an API call of a software development kit (SDK). The method may include receiving a tagged expression indicating that a translation of the software expression from a user API call to an API call of an SDK is to be performed, the SDK being associated with a cloud-native high-performance computing environment; processing an abstract syntax tree associated with the software expression, the processing including replacing symbols in the abstract syntax tree with respective variables, replacing a return statement in the abstract syntax tree with a serialization instruction to write a result to local storage, and serializing the processed abstract syntax tree; and providing the serialized abstract syntax tree and one or more resource files to the cloud-native high-performance computing environment for execution. (A2) In some examples of A1, the method includes creating batch job to execute at least a portion of the serialized abstract syntax tree in the cloud-native high-performance computing environment; creating a storage container associated with the batch job; and creating a specialized resource container associated with the one or more resource files. (A3) In some examples of A1-A2, the batch job is a multitask batch job. (A4) In some examples of A1-A3, the method includes serializing output associated with the return statement to a filename; and adding the filename as an output resource in the one or more resource files. (A5) In some examples of A1-A4, the method includes expanding the software expression by collecting and analyzing symbols in the abstract syntax tree; splitting the software expression into a plurality of individual tasks such that a single expression is created for each task; and performing the plurality of individual tasks in parallel. (A6) In some examples of A1-A5, the method includes analyzing a plurality of software expressions; and automatically tagging at least one of the plurality of software expression resulting in the tagged expression. (A7) In some examples of A1-A6, the method includes providing the serialized abstract syntax tree to the cloud-native high-performance computing environment a resource file.

(70) In yet another aspect, some examples include a system including one or more processors and memory coupled to the one or more processors, the memory storing one or more instructions which when executed by the one or more processors, causes the one or more processors to perform any of the methods described herein (e.g., A1-A7 described above).

(71) In yet another aspect, some examples include a computer-readable storage medium storing one or more programs for execution by one or more processors of a device, the one or more programs including instructions for performing any of the methods described herein (e.g., A1-A7 described above).

(72) The description and illustration of one or more aspects provided in this application are not intended to limit or restrict the scope of the disclosure as claimed in any way. The aspects, examples, and details provided in this application are considered sufficient to convey possession and enable others to make and use the best mode of the claimed disclosure. The claimed disclosure should not be construed as being limited to any aspect, example, or detail provided in this application. Regardless of whether shown and described in combination or separately, the various features (both structural and methodological) are intended to be selectively included or omitted to produce an embodiment with a particular set of features. Having been provided with the description and illustration of the present application, one skilled in the art may envision variations, modifications, and alternate aspects falling within the spirit of the broader aspects of the general

inventive concept embodied in this application that does not depart from the broader scope of the claimed disclosure.

Claims

1. A method comprising: receiving a tagged expression indicating that a translation of a software expression from a user application programming interface (API) call to a cloud API call is to be performed; identifying a function with a return statement in the software expression; replacing the return statement with a serialization step that writes to a file; obtaining an abstract syntax tree associated with the software expression; processing the abstract syntax tree, including replacing symbols in the abstract syntax tree with respective variables; serializing the processed abstract syntax tree; and automatically creating a resource binding, wherein the resource binding uploads the file to a cloud storage area available to a cloud-native high-performance computing environment via a network.
2. The method of claim 1, wherein the cloud API call is an API call of a software development kit (SDK) associated with the cloud-native high-performance computing environment.
3. The method of claim 1, further comprising: providing the serialized processed abstract syntax tree as a resource file to the cloud-native high-performance computing environment for execution.
4. The method of claim 3, further comprising: serializing an output associated with the return statement to a filename; and adding the filename as an output resource in the resource file.
5. The method of claim 1, further comprising: creating a batch job to execute at least a portion of the serialized processed abstract syntax tree in the cloud-native high-performance computing environment; and creating a storage container associated with the batch job.
6. The method of claim 1, further comprising: expanding the software expression by collecting and analyzing the symbols in the abstract syntax tree; splitting the software expression into a plurality of individual tasks such that a single expression is created for each task; and performing the plurality of individual tasks in parallel.
7. The method of claim 1, wherein an individual abstract syntax tree is obtained for each tagged expression.
8. One or more computer storage media including instructions that, when executed by a processor, cause the processor to: receive a tagged expression indicating that a translation of a software expression from a user application programming interface (API) call to a cloud API call is to be performed; identify a function with a return statement in the software expression; replace the return statement with a serialization step that writes to a file; obtaining an abstract syntax tree associated with the software expression; processing the abstract syntax tree, including replacing symbols in the abstract syntax tree with respective variables; serializing the processed abstract syntax tree; and automatically create a resource binding, wherein the resource binding uploads the file to a cloud storage area available to a cloud-native high-performance computing environment via a network.
9. The one or more computer storage media of claim 8, wherein the cloud API call is an API call of a software development kit (SDK) associated with the cloud-native high-performance computing environment.
10. The one or more computer storage media of claim 8, further comprising instructions that, when executed by the processor, cause the processor to: provide the serialized processed abstract syntax tree as a resource file to the cloud-native high-performance computing environment for execution.
11. The one or more computer storage media of claim 8, further comprising instructions that, when executed by the processor, cause the processor to: create a batch job to execute at least a portion of the serialized processed abstract syntax tree in the cloud-native high-performance computing environment; and create a storage container associated with the batch job.
12. The one or more computer storage media of claim 8, further comprising instructions that, when executed by the processor, cause the processor to: expand the software expression by collecting and

analyzing symbols in the abstract syntax tree; split the software expression into a plurality of individual tasks such that a single expression is created for each task; and perform the plurality of individual tasks in parallel.

13. A system comprising: one or more hardware processors; and a user application programming interface (API) and a code generator implemented on the one or more hardware processors and configured by machine-readable instructions to: receive a tagged expression indicating that a translation of a software expression from a user API call to a cloud API call is to be performed; identify a function with a return statement in the software expression; replace the return statement with a serialization step that writes to a file; obtaining an abstract syntax tree associated with the software expression; processing the abstract syntax tree, including replacing symbols in the abstract syntax tree with respective variables; serializing the processed abstract syntax tree; and automatically create a resource binding, wherein the resource binding uploads the file to a cloud storage area available to a cloud-native high-performance computing environment via a network.

14. The system of claim 13, wherein the user API and the code generator is further configured to: provide the serialized processed abstract syntax tree as a resource file to the cloud-native high-performance computing environment for execution.

15. The system of claim 13, wherein the user API and the code generator is further configured to: create a batch job to execute at least a portion of the serialized processed abstract syntax tree in the cloud-native high-performance computing environment; and create a storage container associated with the batch job.

16. The system of claim 13, wherein the user API and code generator is further configured to: expand the software expression by collecting and analyzing the symbols in the abstract syntax tree; split the software expression into a plurality of individual tasks such that a single expression is created for each task; and perform the plurality of individual tasks in parallel.

17. The system of claim 13, wherein the cloud API call is an API call of a software development kit (SDK) associated with the cloud-native high-performance computing environment.
